INCH-POUND

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21 February 1985

MILITARY SPECIFICATION

BADGE, RATING AND QUALIFICATION, DUNGAREE UNIFORM

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an iron-on rating and qualification badge for dungaree uniforms.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, P.O. Box 59, Natick, MA 01760-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8455

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SPECIFICATIONS

MILITARY

* MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-B-17757	- Box, Shipping, Fiberboard (Modular Sizes)
MIL-C-87065	- Cloth, Chambray, Polyester and Cotton

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
* MIL-STD-2073-1	- DOD Materials Procedures for Development and Application of Packaging Requirements
* MIL-STD-2073-2	- Packaging Requirement Codes

* (Unless otherwise indicated, copies of federal and military specification, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Applications for copies should be addressed to the American Trucking Association, ATTN: Traffic Department, 1616 P Street, N.W., Washington, DC 20036-1404.)

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UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606-5808.)

* (Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample in which case this specification shall govern.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Basic material. The basic material shall be a polyester/cotton chambray cloth conforming to MIL-C-87065, except the seam efficiency requirements specified therein shall not apply.

3.3.2 Coating compound. The coating compound used to adhere the badge shall be a clear thermoplastic resin suitably compounded to meet the requirements specified herein. The resin shall be bacteria and fungi resistant, non-toxic, non-irritant, and free of any impurity that might cause fabric deterioration or discoloration on the printed side of the finished badge (see 4.4.1).

3.3.3 Printing ink. The ink for printing the insignia on face side of badge shall be of a suitable black shade to match the guide sample.

3.3.3.1 Colorfastness. The printed matter on the finished badge shall show a minimum of "good" fastness to light, hot-dry pressing, chlorine bleaching and laundering when tested as specified in 4.5.

3.4 Coated fabric. The basic chambray material shall be evenly coated on one side only with the resin compound specified in 3.3.2, to finish with the coating heat sealable with good adhesion characteristics. The coated side shall not show any evidence of blocking when tested as specified in 4.4.1. An acceptable degree of adhesion on the coated side of the finished badge shall be determined as follows:

No clear evidence of curling or peeling will result when the badge is subjected to mobile launderings and accelerated aging as specified in 4.5.

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3.5 Figures. Figures are furnished for information purposes only. When inconsistancies exist between the written specification and the figures, the written specification shall govern.

3.6 Workmanship. The finished badges shall conform to the quality established by this specification. The occurrance of defects shall not exceed the applicable acceptable quality levels.

3.7 Tolerances. The tolerances shown in Table I shall apply for the dimensions as indicated. All measurements are in inches (see 4.4.2.2 and figures).

Table I - Tolerances

Dimension	Tolerance
Any specified dimension under 15/16	± 1/32
Any specified dimension of 1 and up to 1-15/16	± 1/16
Any specified dimension of 2 and up to 2-15/16	± 3/32
Any specified dimension of 3 and up to 3-15/16	± 1/8
Any specified dimension of 4 and up to 4-15/16	± 5/32
Any specified dimension of 5 and up to 5-15/16	± 3/16
Any specified dimension of 6 and up to 6-15/16	± 7/32
Any specified dimension of 7 and up to 7-15/16	± 1/4

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.4.2 for compliance with design, construction, workmanship, appearance and dimensional requirements.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, components and materials listed in Table II shall be tested for the characteristics specified and in accordance with FED-STD-191 whenever applicable. All test reports shall contain individual values utilized in expressing the final result. The lot size for the coated fabric shall be expressed in yards and the sample unit shall be 1/2 yard (full width). The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirement specified.

Table II - Component testing

Component	Characteristic	Requirement paragraph	Test method
Coating compound	Resistance to Bacteria and Fungi	3.3.2	1/
	Material Identification	3.3.2	1/
	Toxicity	3.3.2	1/
Cloth, coated	Coating, heat sealable	3.4	1/
	Blocking	3.4	5872

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement. In respect to the toxicity of the coating, the contractor must confirm that irritants or chemicals that could cause dermatitis problems or deterioration of the basic fabric while in storage are not present.

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4.4.2 Examination of the end item. The defects found during the examination of the end item shall be classified in accordance with 4.4.2.1 and 4.4.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.4.2.3. The lot size shall be expressed in units of one badge. The sample unit shall be one finished badge.

4.4.2.1 Visual examination.

Examine	Defect
Design	Any deviation from illustration or guide sample. Color of printing not as specified. Design poorly defined, distorted, or misshaped. Not properly positioned or centered.
Material	Any woven or finishing defect seriously affecting the overall appearance of the badge. Any hole, cut, or tear. Any uncoated area clearly visible on back side of badge. Coating omitted.
Workmanship	Spot, stain, or imbedded foreign matter clearly noticeable on face side of badge. Print not distinct or blurred Overall discoloration clearly noticeable on face side of badge.

4.4.2.2 Dimensional examination. The finished badge shall be examined for dimensional defects in accordance with the applicable figure. Any measurement deviating from the dimension specified shall be scored as a size measurement defect (see Table I for tolerance).

4.4.2.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels, expressed in defects per 100 units for visual and dimensional examination, shall be as follows:

	Inspection level	AQL
For defects applicable to 4.4.2.1	II	4.0
For defects applicable to 4.4.2.2	S-3	6.5

4.4.3 Examination of packaging requirements. An examination shall be made to determine that packaging, packing, and marking comply with Section 5 requirements of this specification. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be on the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 2.5, expressed in terms of defects per hundred units.

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, improper taping, inadequate stapling, bulged or distorted container.
Content	Number of items per shipping container is more or less than required. Size shown on one or more items not as specified on shipping container. 1/

1/ For this defect, one item from each shipping container in sample shall be examined.

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5 as specified in MIL-STD-105.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

4.5 Testing of the end item. Each lot of the end item shall be tested for the characteristics specified in Table III and in accordance with FED-STD-191 whenever applicable. The physical and chemical values specified in Section 3 apply to the average of the determinations made on the sample unit for test purposes as specified in the applicable test method. The lot size shall be expressed in units of one badge. The sample unit for all tests shall be seven (7) badges. The inspection level shall be S-1 with an AQL of 6.5 defects per 100 units. All test reports shall contain the individual values utilized in expressing the final results.

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Table III - End item tests

Characteristic	Requirement Paragraph	Test Method
Colorfastness to:		
Light	3.3.3.1	5660
Dry pressing only	3.3.3.1	5640
Chlorine bleaching	3.3.3.1	5600
Laundering	3.3.3.1	5519
Adhesion of coating	3.4	4.5.1
Mobile laundering	3.4	5556 1/ and 4.5.2
Accelerated aging	3.4	5850 2/ and 4.5.3

1/ Cotton laundering procedure only shall be applicable.

2/ Except that the test specimen shall be tested at a temperature of 160° (± 4°)F.

4.5.1 Adhesion of coating. The test specimen shall be prepared as follows:

- Position and center a badge (coated side down) to an approximate 8 by 8 inch specimen of the polyester/cotton chambray material specified in 3.3.1, with the warp direction of the respective component parts parallel to each other.
- Press the badge firmly to the chambray material with a dry hot iron for a minimum of 20 seconds, moving the iron across its center and around the edges.
- The temperature of the dry hot iron shall be approximately 350°F or with the selection placed on the "cotton" setting.
- Allow sufficient time for cooling.

One determination shall be performed per sample unit for evidence of poor adhesion, i.e., curling or peeling from the basic material. The result shall be reported as "pass" or "fail".

4.5.2 Mobile laundering. A test specimen prepared as specified in 4.5.1, shall be subjected to five (5) launderings maximum and evaluated at each sequence. The edges of the badge shall be examined for distinct evidence of curling or peeling, and shall be reported as "pass" or "fail". The flexibility and stiffness of the badge after each laundering must be maintained at least equal to the original sample before laundering.

4.5.3 Accelerated aging. The test specimen shall be prepared as specified in 4.5.1. One determination shall be performed per sample unit for distinct evidence of deterioration. The result shall be as "pass" or "fail".

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or C, as specified (see 6.2).

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5.1.1 Level A. Twelve (12) rating badges or five (5) qualification badges stacked neatly in a bundle, shall be inserted face side up into a snug-fitting, flat style, clear polyethylene film bag of 0.002 inch thickness (minimum). The bag may be fabricated from polyethylene film tubing or sheeting. The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed, with the heat seal made as close as possible to the open end. A 1/4 inch diameter hole shall be made at one corner of the polyethylene bag to allow excess air to escape. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a tuck or reverse flap.

5.1.2 Level C. The badges shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The contractor may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C (see 6.2) for rating badges. Packing shall be level C for qualification badges.

5.2.1 Level A. Two hundred fifty two (252) bundles of rating badges, preserved as specified in 5.1, shall be packed in a corrugated fiberboard shipping container conforming to type CF, class weather-resistant, size 1A, variety DW, grade V15c of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Closure and reinforcement requirements shall be as specified in the appendix of the box specification. Level A packages shall be packed flat, three in length, three in width, and twenty eight in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.2 Level B. Two hundred fifty two (252) bundles of rating badges, preserved as specified in 5.1, shall be packed in a corrugated fiberboard shipping container conforming to type CF, class domestic, size 1A, variety DW, grade 200 of MIL-B-17757. The fiberboard for the box liner shall conform to type CF, class domestic of MIL-B-17757. Closure and reinforcement requirements shall be as specified in the appendix of the box specification. Level A packages shall be packed flat, three in length, three in width, and twenty eight in depth. Each container shall have the contents completely covered on the top and bottom with a sheet of commercial grade kraft paper.

5.2.3 Level C. Items preserved, as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with the US Postal Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Marking. In addition to any special marking required by the contract or order, unit packs, intermediate or exterior containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

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5.3.1 Polyethylene bagged packages. Polyethylene bagged packages shall have the following information legibly printed or stamped in black directly on the bag across the center face or on a white paper label inserted within the bag, so to permit ready identification.

INSTRUCTIONS FOR APPLYING BADGE

DO NOT USE STEAM IRON

SET INDICATOR ON DRY IRON AT 350°F OR "COTTON". ALLOW IRON TO GET COMPLETELY HEATED. PRESS WRINKLES OUT OF GARMENT. POSITION BADGE ON GARMENT WITH GLOSSY SIDE DOWN. PRESS IRON DOWN HARD AND MOVE ACROSS CENTER AND AROUND EDGES FOR A MINIMUM OF 20 SECONDS. ALLOW TO COOL. IF ANY EDGE CURLS, PRESS AREA AGAIN. TO PREVENT SCORCHING, PLACE A PIECE OF HEAVY PAPER OR CLOTH BETWEEN IRON AND BADGE WHEN PRESSING.

* 5.4 Palletization. When specified (see 6.2), item, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the bonding methods C and D or Film bonding F or G. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type IV; type V, class 1, size 2; or type VIII, fabricated from wood groups I, II, III or IV, grade A of NN-P-71, or 4-way, style 1, size A, type I, class 1 fabricated from groups specified of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

* 5.5 MIL-STD-2073. When specified (see 6.2), Preservation and Packing shall be in accordance with MIL-STD-2073-1 and MIL-STD-2073-2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The badges are intended for use by Naval enlisted personnel. The First Class Petty Officer Badge (see Figure 1) can be cut for use by personnel to accomodate Second and Third Class Petty Officers.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Badge idenfication
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
d. When first article sample is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quality, testing and approval of the first article.
e. Levels of preservation and packing required (see 5.1 and 5.2)
f. When palletization is required (see 5.4)
g. When all packaging data is to be found in MIL-STD-2073-1 and MIL-STD-2073-2 (see 5.5).

6.3 Samples. For access to guide samples, address the procuring activity issuing the invitation for bids (see 3.1).

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

* 6.5 Subject term (key word) listing.

Cloth, chambray
Crow, iron-on

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

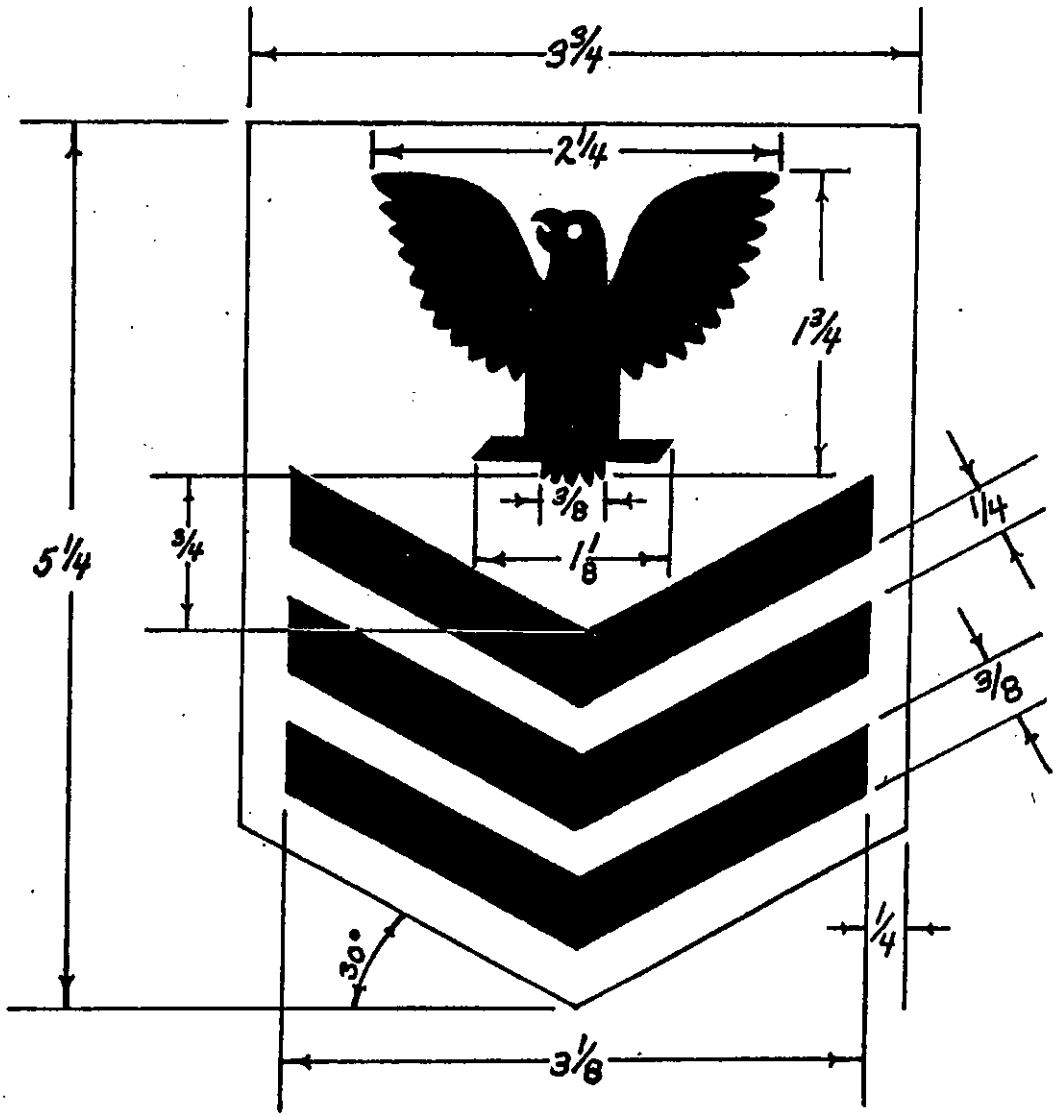
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DLA - CT

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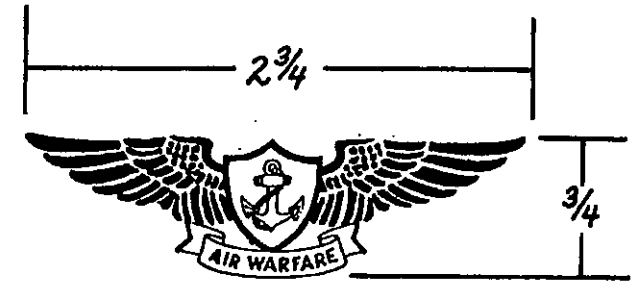
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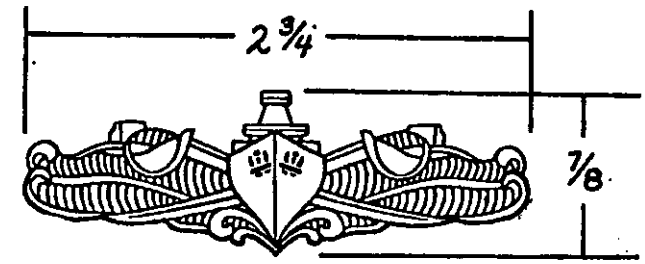
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FIGURE 1 - BADGE, RATING, DUNGAREE

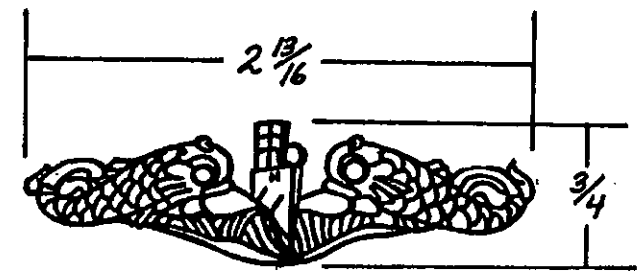
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AVIATION WARFARE SPECIALIST



ENLISTED SURFACE WARFARE SPECIALIST



SUBMARINE

NOTE: SIZE OF THE FINISHED CHAMBRAY PATCH SHALL BE 3 1/4 X 1 1/4 INCHES

FIG. 2 - BADGE, QUALIFICATION, DUNGAREE

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